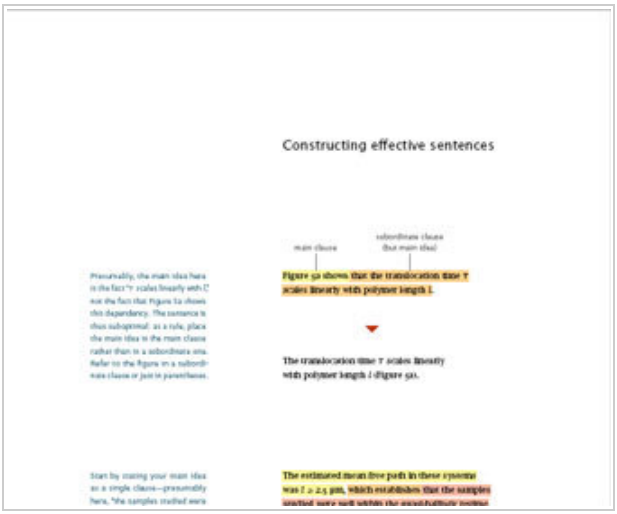
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CONCEPT

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Effective Writing

Effective writing is readable — that is, clear, accurate, and concise. When you are writing a paper, try to get your ideas across in such a way that the audience will understand them effortlessly, unambiguously, and rapidly. To this end, strive to write in a straightforward way. There is no need to write about science in unusual, complicated, or overly formal ways in an effort to "sound scientific" or to impress your audience. If you can tell a friend about your work, you are off to a good start.



Constructing effective sentences

The six sentences shown here lack impact because they place their main idea in a subordinate clause. They are best rewritten in a simpler way.

To construct sentences that reflect your ideas, focus these sentences appropriately. Express one idea per sentence. Use your current topic — that is, what you are writing about — as the grammatical subject of your sentence (see *Verbs: Choosing between active and passive voice*). When writing a complex sentence (a sentence that includes several clauses), place the main idea in the main clause rather than a subordinate clause. In particular, focus on the phenomenon at hand, not on the fact that you observed it.

Constructing your sentences logically is a good start, but it may not be enough. To ensure they are readable, make sure your sentences do not tax readers' short-term memory by obliging these readers to remember long pieces of text before knowing what to do with them. In other words, keep together what goes together. Then, work on conciseness: See whether you can replace long phrases with shorter ones or eliminate words without loss of clarity or accuracy.

The following screens cover the drafting process in more detail. Specifically, they discuss how to use verbs effectively and how to take care of your text's mechanics.

Verbs



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Much of the strength of a clause comes from its verb. Therefore, to express your ideas accurately, choose an appropriate verb and use it well. In particular, use it in the right tense, choose carefully between active and passive voice, and avoid dangling verb forms.

Verbs are for describing actions, states, or occurrences. To give a clause its full strength and keep it short, do not bury the action, state, or occurrence in a noun (typically combined with a weak verb), as in "The catalyst produced a significant increase in conversion rate." Instead write, "The catalyst increased the conversion rate significantly." The examples below show how an action, state, or occurrence can be moved from a noun back to a verb.

Instead of	*Write*
Make an examination of . . .	examine
Present a comparison of . . .	compare
Be in agreement . . .	agree
Perform an analysis of . . .	analyze
Produce an improvement in . . .	improve

Using the right tense

In your scientific paper, use verb tenses (past, present, and future) exactly as you would in ordinary writing. Use the past tense to report what happened in the past: what you did, what someone reported, what happened in an experiment, and so on. Use the present tense to express general truths, such as conclusions (drawn by you or by others) and atemporal facts (including information about what the paper does or covers). Reserve the future tense for perspectives: what you will do in the coming months or years. Typically, most of your sentences will be in the past tense, some will be in the present tense, and very few, if any, will be in the future tense.

Past tense

Work done

We collected blood samples from . . .
Groves et al. determined the growth rate of . . .
Consequently, astronomers decided to rename . . .

Work reported
Jankowsky reported a similar growth rate . . .
In 2009, Chu published an alternative method to . . .
Irarrázaval observed the opposite behavior in . . .

Observations
The mice in Group A developed, on average, twice as much . . .
The number of defects increased sharply . . .
The conversion rate was close to 95% . . .

Present tense

General truths
Microbes in the human gut have a profound influence on . . .
The Reynolds number provides a measure of . . .
Smoking increases the risk of coronary heart disease . . .

Atemporal facts
This paper presents the results of . . .
Section 3.1 explains the difference between . . .
Behbood's 1969 paper provides a framework for . . .

Future tense

Perspectives
In a follow-up experiment, we will study the role of . . .
The influence of temperature will be the object of future research . . .

Note the difference in scope between a statement in the past tense and the same statement in the present tense: "The temperature increased linearly over time" refers to a specific experiment, whereas "The temperature increases linearly over time" generalizes the experimental observation, suggesting that the temperature always increases linearly over time in such circumstances.

In complex sentences, you may have to combine two different tenses — for example, "In 1905, Albert Einstein postulated that the speed of light is constant . . . . " In this sentence, *postulated* refers to something that happened in the past (in 1905) and is therefore in the past tense, whereas *is* expresses a general truth and is in the present tense.

Choosing between active and passive voice
In English, verbs can express an action in one of two voices. The active voice

focuses on the agent: "John measured the temperature." (Here, the agent — John — is the grammatical subject of the sentence.) In contrast, the passive voice focuses on the object that is acted upon: "The temperature was measured by John." (Here, the temperature, not John, is the grammatical subject of the sentence.)

To choose between active and passive voice, consider above all what you are discussing (your topic) and place it in the subject position. For example, should you write "The preprocessor sorts the two arrays" or "The two arrays are sorted by the preprocessor"? If you are discussing the preprocessor, the first sentence is the better option. In contrast, if you are discussing the arrays, the second sentence is better. If you are unsure what you are discussing, consider the surrounding sentences: Are they about the preprocessor or the two arrays?

surrounding

The desire to be objective in scientific writing has led to an overuse of the passive voice, often accompanied by the exclusion of agents: "The temperature was measured" (with the verb at the end of the sentence). Admittedly, the agent is often irrelevant: No matter who measured the temperature, we would expect its value to be the same. However, a systematic preference for the passive voice is by no means optimal, for at least two reasons.

don't overuse passive

For one, sentences written in the passive voice are often less interesting or more difficult to read than those written in the active voice. A verb in the active voice does not require a person as the agent; an inanimate object is often appropriate. For example, the rather uninteresting sentence "The temperature was measured . . . " may be replaced by the more interesting "The measured temperature of 253°C suggests a secondary reaction in . . . ." In the second sentence, the subject is still *temperature* (so the focus remains the same), but the verb *suggests* is in the active voice. Similarly, the hard-to-read sentence "In this section, a discussion of the influence of the recirculating-water temperature on the conversion rate of . . . is presented" (long subject, verb at the end) can be turned into "This section discusses the influence of . . . . " The subject is now *section*, which is what this sentence is really about, yet the focus on the discussion has been maintained through the active-voice verb *discusses*.

make it the subject

no passive

As a second argument against a systematic preference for the passive voice, readers sometimes need people to be mentioned. A sentence such as "The temperature is believed to be the cause for . . . " is ambiguous. Readers will want to know *who* believes this — the authors of the paper, or the scientific community as a whole? To clarify the sentence, use the active voice and set the appropriate people as the subject, in either the third or the first person, as in the examples below.

> Biologists believe the temperature to be . . .
> Keustermans et al. (1997) believe the temperature to be . . .
> The authors believe the temperature to be . . .

We believe the temperature to be . . .

Avoiding dangling verb forms

A verb form needs a subject, either expressed or implied. When the verb is in a non-finite form, such as an infinitive (*to do*) or a participle (*doing*), its subject is implied to be the subject of the clause, or sometimes the closest noun phrase. In such cases, construct your sentences carefully to avoid suggesting nonsense. Consider the following two examples.

To dissect its brain, the affected fly was mounted on a . . .
After aging for 72 hours at 50°C, we observed a shift in . . .

Here, the first sentence implies that the affected fly dissected its own brain, and the second implies that the authors of the paper needed to age for 72 hours at 50°C in order to observe the shift. To restore the intended meaning while keeping the infinitive *to dissect* or the participle *aging*, change the subject of each sentence as appropriate:

To dissect its brain, we mounted the affected fly on a . . .
After aging for 72 hours at 50°C, the samples exhibited a shift in . . .

Alternatively, you can change or remove the infinitive or participle to restore the intended meaning:

To have its brain dissected, the affected fly was mounted on a . . .
After the samples aged for 72 hours at 50°C, we observed a shift in . . .

Mechanics

In communication, every detail counts. Although your focus should be on conveying your message through an appropriate structure at all levels, you should also save some time to attend to the more mechanical aspects of writing in English, such as using abbreviations, writing numbers, capitalizing words, using hyphens when needed, and punctuating your text correctly.

Using abbreviations

Beware of overusing abbreviations, especially acronyms — such as *GNP* for *gold nanoparticles*. Abbreviations help keep a text concise, but they can also render it cryptic. Many acronyms also have several possible extensions (*GNP* also stands for *gross national product*).

Write acronyms (and only acronyms) in all uppercase (*GNP*, not *gnp*).

Introduce acronyms systematically the first time they are used in a document. First write the full expression, then provide the acronym in parentheses. In the full expression, and unless the journal to which you submit your paper uses a different convention, capitalize the letters that form the acronym: "we prepared Gold NanoParticles (GNP) by . . . " These capitals help readers quickly recognize

what the acronym designates.

Notes:

- Do not use capitals in the full expression when you are not introducing an acronym: "we prepared gold nanoparticles by… "
- As a more general rule, use first what readers know or can understand best, then put in parentheses what may be new to them. If the acronym is better known than the full expression, as may be the case for techniques such as SEM or projects such as FALCON, consider placing the acronym first: "The FALCON (Fission-Activated Laser Concept) program at…"
- In the rare case that an acronym is commonly known, you might not need to introduce it. One example is DNA in the life sciences. When in doubt, however, introduce the acronym.

In papers, consider the abstract as a stand-alone document. Therefore, if you use an acronym in both the abstract and the corresponding full paper, introduce that acronym twice: the first time you use it in the abstract and the first time you use it in the full paper. However, if you find that you use an acronym only once or twice after introducing it in your abstract, the benefit of it is limited — consider avoiding the acronym and using the full expression each time (unless you think some readers know the acronym better than the full expression).

Writing numbers

In general, write single-digit numbers (zero to nine) in words, as in *three hours*, and multidigit numbers (10 and above) in numerals, as in *24 hours*. This rule has many exceptions, but most of them are reasonably intuitive, as shown hereafter.

Use numerals for numbers from zero to nine

- when using them with abbreviated units (*3 mV*);
- in dates and times (*3 October*, *3 pm*);
- to identify figures and other items (*Figure 3*);
- for consistency when these numbers are mixed with larger numbers (*series of 3, 7, and 24 experiments*).

Use words for numbers above 10 if these numbers come at the beginning of a sentence or heading ("Two thousand eight was a challenging year for . . . "). As an alternative, rephrase the sentence to avoid this issue altogether ("The year 2008 was challenging for . . . ").

Capitalizing words

Capitals are often overused. In English, use initial capitals

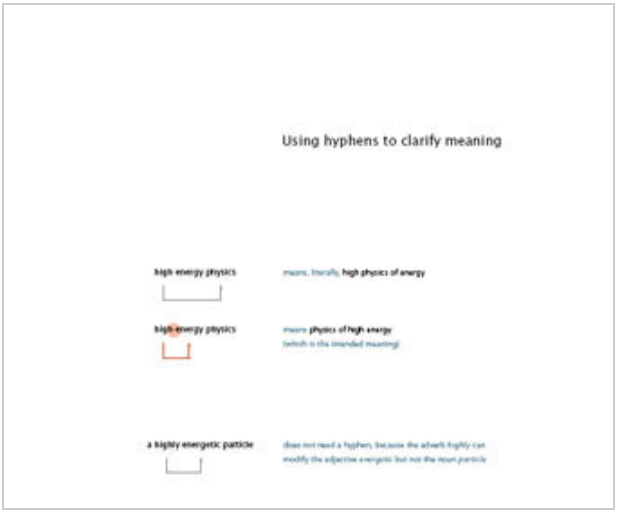
- at beginnings: the start of a sentence, of a heading, etc.;
- for proper nouns, including nouns describing groups (compare *physics* and *the Physics Department*);

- for items identified by their number (compare *in the next figure* and *in Figure 2*), unless the journal to which you submit your paper uses a different convention;
- for specific words: names of days (*Monday*) and months (*April*), adjectives of nationality (*Algerian*), etc.

In contrast, do not use initial capitals for common nouns: Resist the temptation to glorify a concept, technique, or compound with capitals. For example, write *finite-element method* (not *Finite-Element Method*), *mass spectrometry* (not *Mass Spectrometry*), *carbon dioxide* (not *Carbon Dioxide*), and so on, unless you are introducing an acronym (see *Mechanics: Using abbreviations*).

Using hyphens

Use hyphens in English to clarify relationships in chains of words. Thus, *low temperature impact* (without a hyphen) suggests a low impact of the temperature, whereas *low-temperature impact* (with a hyphen) suggests the impact of or at low temperature. Such hyphens, useful for (nouns used as) adjectives, are unnecessary for adverbs. For example, *a highly interesting paper* does not need a hyphen; in this phrase, *highly* can only qualify *interesting* (not *paper*).



Using hyphens to clarify meaning

Hyphens can be tricky in English, especially in long expressions. Here is a more detailed example.

In general, do not use a hyphen with a prefix, namely an element that is not a word in itself and that is added at the beginning of a word to modify its meaning. Thus, write *multichannel*, *nonlinear*, *preamplifier*, *postdoctoral*, *realign*, etc. As an exception to this rule, use a hyphen to separate vowels that would otherwise be read together, as in *pre-embryo*, or when the original word is written with a capital, as in *pre-Columbian*.

Punctuating text

Punctuation has many rules in English; here are three that are often a challenge for non-native speakers.

As a rule, insert a comma between the subject of the main clause and whatever comes in front of it, no matter how short, as in "Surprisingly, the temperature did not increase." This comma is not always required, but it often helps and never hurts the meaning of a sentence, so it is good practice.

In series of three or more items, separate items with commas (*red, white, and blue*;

yesterday, today, or tomorrow). Do not use a comma for a series of two items (*black and white*).

In displayed lists, use the same punctuation as you would in normal text (but consider dropping the *and*).

> The system is fast, flexible, and reliable.
>
> The system is
>
> - fast,
> - flexible,
> - reliable.

eBooks

This page appears in the following eBook

English Communication for Scientists, Unit 2.2